

The Minimal Degree of a 2-Primary Brauer Jump on the Fourth Hénon Yukawa Surface

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August 2026

Abstract

We determine the first finite base changes at which the algebraic Brauer quotient of a fixed smooth cubic surface can acquire 2-primary torsion. The normal field of the surface's twenty-seven lines has Galois group $W(E_6)$, and for every finite extension $L/\mathbf{Q}$ a nonzero 2-primary quotient forces the sharp divisibility

$$36 \mid [K \cap L : \mathbf{Q}] \mid [L : \mathbf{Q}].$$

Equality occurs exactly at the thirty-six conjugate fixed fields of double-six stabilizers; for a double-six D , the attaining field $F_D = K^{U_1}$, where $U_1 \cong S_6 \times C_2$, has an oriented quadratic extension $F_D(\sqrt{\delta_D})$. We construct a unique normalized quartic Q_D by a determinant in a 60×31 exact restriction matrix and prove $\text{div}(Q_D) = \mathcal{E} + \mathcal{G}$, the sum of the twelve lines of D . The quaternion $(\delta_D, Q_D/u_0^4)$ is then unramified and represents the unique nonzero element of $\text{Br}(Y_{F_D})/\text{im Br}(F_D)$. The exact evidence enumerates thirty-six double-sixes, proves a rank-30 restriction theorem, and passes an independent theorem checker importing no producer theorem helper. The bounded 2026-08-15 screen did not locate a prior exact computation of this field-orientation-quaternion package for the fixed surface; generic resolvers and double-six Brauer classes are prior constructions rather than contributions of this paper. We make no local-evaluation, rational-point, or Brauer–Manin claim.

1 Introduction

For a smooth cubic surface whose twenty-seven lines have the full incidence-compatible Galois group $W(E_6)$, the algebraic Brauer quotient may be trivial over the base yet acquire 2-torsion after a finite extension. How small can that extension be? We answer this for the exact surface $Y/\mathbf{Q}$ in equation (3). The companion theorem package `henon_mu3_yukawa_line_field`, included in the anonymous supplement, proves that the common normal field K of its lines satisfies

$$\text{Gal}(K/\mathbf{Q}) \cong W(E_6), \quad |W(E_6)| = 51840.$$

The degree-27 field E of one line is non-Galois and differs from K . These are frozen inputs: we locate the first jump inside K , not recompute the line scheme.

Nonzero 2-primary cohomology has two classical branches. A quotient $\mathbf{Z}/2$ places the Galois image in a conjugate of the double-six stabilizer $U_1 \cong S_6 \times C_2$, of index 36; a quotient $(\mathbf{Z}/2)^2$ places it in a conjugate of U_3 , of index 720. Finding one invariant double-six does not eliminate the second branch. The complete Swinnerton-Dyer–Elsenhans–Jahnel classification supplies this universal dichotomy [5, 2].

The result combines that classification with exact configuration arithmetic and a written descent argument. For an abelian group A , write $A[2]$ for its subgroup annihilated by 2. For a finite extension $L/\mathbf{Q}$, set

$$G_L = \text{Gal}(\overline{\mathbf{Q}}/L), \quad H_L = \text{im}(G_L \rightarrow W(E_6)), \quad N_L = \ker(G_L \rightarrow H_L). \quad (1)$$

The cohomological passage from G_L to H_L is proved in lemma 5.1 and equation (33), applied with $k = L$.

Theorem 1.1 (Main theorem). *Let $Y/\mathbf{Q}$ be the cubic surface in equation (3), and let $K/\mathbf{Q}$ be the common normal field of its twenty-seven lines. For every finite extension $L/\mathbf{Q}$,*

$$(\mathrm{Br}(Y_L)/\mathrm{im}\mathrm{Br}(L))[2] \neq 0 \implies 36 \mid [K \cap L : \mathbf{Q}] \mid [L : \mathbf{Q}]. \quad (2)$$

Consequently the least possible degree is 36. If $[L : \mathbf{Q}] = 36$ and the left side of equation (2) is nonzero, then, inside the fixed algebraic closure,

$$L = K \cap L = K^{wU_1w^{-1}}$$

for some $w \in W(E_6)$, and this subgroup stabilizes a unique embedded double-six. The thirty-six embedded fields so obtained form one $\mathbf{Q}$ -isomorphism type.

For a selected double-six $D = \{\mathcal{E}, \mathcal{G}\}$, put $F_D = K^{U_1}$. There are exact invariants $\theta_D, \delta_D \in K$ such that

$$\mathbf{Q}(\theta_D) = F_D = \mathbf{Q}(\delta_D), \quad F'_D = K^{U_1^+} = F_D(\sqrt{\delta_D}),$$

where $U_1^+ \cong S_6$ preserves the two sixers separately. Moreover,

$$\mathrm{Br}(Y_{\mathbf{Q}})/\mathrm{im}\mathrm{Br}(\mathbf{Q}) = 0, \quad \mathrm{Br}(Y_{F_D})/\mathrm{im}\mathrm{Br}(F_D) \cong \mathbf{Z}/2.$$

There is a unique normalized determinant-defined quartic Q_D satisfying

$$\mathrm{div}_{Y_{F_D}}(Q_D) = \mathcal{E} + \mathcal{G},$$

and the quaternion

$$\mathcal{A}_D = (\delta_D, Q_D/u_0^4)$$

is unramified and represents the unique nonzero element of $\mathrm{Br}(Y_{F_D})/\mathrm{im}\mathrm{Br}(F_D)$.

The contributions are:

1. the full divisibility chain and equality classification through both U_1/U_3 branches;
2. exact degree-36 fields, their common normal closure K , and the oriented quadratic extension;
3. a determinant quartic whose degree-exhausted divisor and integral Picard cocycle prove that $(\delta_D, Q_D/u_0^4)$ is unramified and nonzero.

Invariant double-sixes, orientation fields, and cyclic algebras are established mechanisms [2, 1]; degree-36 resolvers and their moduli context are also prior art [3, 4]. Our claim is instance-specific: the bounded 2026-08-15 screen did not locate a prior exact computation of the degree-36 double-six field, its orientation square, and the determinant-defined quaternion generator for this frozen Yukawa surface.

The proof is organized around the path visible in figure 1. Section 2 fixes the surface, notation, sources, and evidence contract. Section 3 proves the universal minimum. Sections 4 and 5 identify the fields and cohomological jump. The determinant quartic is constructed in section 6; its divisor and quaternion class are proved in section 7. Exact replay and limitations appear in section 8. We compute no local evaluation and draw no conclusion about rational points, the Hasse principle, weak approximation, or a Brauer–Manin obstruction.

branch	forced Galois image	degree consequence
$\mathbf{Z}/2$	$H_L \subseteq wU_1w^{-1}, [W(E_6) : U_1] = 36$	equality occurs at $F_D = K^{U_1}$
$(\mathbf{Z}/2)^2$	$H_L \subseteq wU_3w^{-1}, [W(E_6) : U_3] = 720$	excluded when $[L : \mathbf{Q}] = 36$

$$K \supset F'_D = K^{U_1^+} = F_D(\sqrt{\delta_D}) \supset F_D = K^{U_1} \supset \mathbf{Q}.$$

$$F_D \implies (\delta_D, Q_D/u_0^4) \in \text{Br}(Y_{F_D}),$$

$$[K \cap L : \mathbf{Q}] = [W(E_6) : H_L], \quad 36 \mid [K \cap L : \mathbf{Q}] \mid [L : \mathbf{Q}].$$

Figure 1: Both classification branches force divisibility by 36, but only U_1 attains equality. Its fixed field, orientation extension, and quaternion form the attaining package; no local evaluation is asserted.

2 Prior results, the frozen object, and the proof contract

2.1 The fixed surface and its line fields

In homogeneous coordinates u_0, u_1, u_2, u_3 , let

$$\begin{aligned}
F = & 75081586157u_0^3 - 28576620789u_0^2u_1 - 122000922135u_0^2u_2 - 5364921951u_0^2u_3 \\
& + 164150208636u_0u_1^2 - 415458334296u_0u_1u_2 + 151070718312u_0u_1u_3 \\
& + 1158143874300u_0u_2^2 + 114691988016u_0u_2u_3 + 113572676646u_0u_3^2 \\
& + 6898957820u_1^3 + 1132596902196u_1^2u_2 - 30413540316u_1^2u_3 \\
& - 2054867641020u_1u_2^2 + 151980984216u_1u_2u_3 + 36794420832u_1u_3^2 \\
& + 2646295985484u_2^3 + 560186573940u_2^2u_3 + 706181383584u_2u_3^2 + 1884468968u_3^3, \quad (3)
\end{aligned}$$

and set $Y = V(F) \subset \mathbf{P}_{\mathbf{Q}}^3$. Its primitive, positive-leading normalization fixes the common scalar. The upstream exact theorem proves that Y is smooth and geometrically irreducible and that its Fano scheme of lines is connected of degree 27. Write $E/\mathbf{Q}$ for that non-Galois residue field and $K/\mathbf{Q}$ for its normal closure. The frozen arithmetic input is

$$[E : \mathbf{Q}] = 27, \quad E \neq K, \quad \text{Gal}(K/\mathbf{Q}) \cong W(E_6), \quad |W(E_6)| = 51840. \quad (4)$$

Thus K is the common normal field of the lines; E is not used as a configuration field.

Let

$$\Lambda = \text{Pic}(Y_{\overline{\mathbf{Q}}})$$

be the rank-seven geometric Picard lattice. In a marked blow-up model we use the basis $h, e_1, \dots, e_6$, with intersection form $\text{diag}(1, -1, \dots, -1)$, and hyperplane class

$$H_Y = 3h - e_{\Sigma}, \quad e_{\Sigma} = e_1 + \dots + e_6. \quad (5)$$

The 27 line classes are e_i , $h - e_i - e_j$, and $2h - \sum_{j \neq i} e_j$. Their intersection graph is the exact Schläfli configuration used throughout. The anonymous supplement names the surface package `henon_mu3_rational_yukawa_surface` and the companion line-field package `henon_mu3_yukawa_line_field`; no external bibliographic claim is attached to those internal package names.

2.2 Double-sixes and fixed fields

A *sixer* is a set of six pairwise skew lines. A double-six is an unordered pair $D = \{\mathcal{E}, \mathcal{G}\}$ of opposite sixers; the same symbols $\mathcal{E}$ and $\mathcal{G}$ denote the corresponding effective sums of six lines. Fix one D , and choose the marking in equation (5) so that

$$[\mathcal{E}] = e_\Sigma, \quad [\mathcal{G}] = 12h - 5e_\Sigma. \quad (6)$$

Define

$$U_1 = \text{Stab}_{W(E_6)}(D) \cong S_6 \times C_2, \quad (7)$$

$$U_1^+ = \text{Stab}_{W(E_6)}(\mathcal{E}) \cap \text{Stab}_{W(E_6)}(\mathcal{G}) \cong S_6, \quad (8)$$

$$F_D = K^{U_1}, \quad F'_D = K^{U_1^+}. \quad (9)$$

Thus U_1^+ has index two in U_1 ; the nontrivial element of the quotient exchanges the two sixers. The other subgroup needed for the universal argument is

$$U_3 \cong (S_3 \times S_3) \rtimes C_2, \quad |U_3| = 72, \quad [W(E_6) : U_3] = 720.$$

It stabilizes a classical triple of azygetic double-sixes. The selected machine model of U_1 will not be used as a substitute for the theorem classifying every subgroup with nonzero 2-primary cohomology. Whenever an S_6 -action appears below, it means this embedded subgroup U_1^+ , not an unspecified abstract copy.

2.3 Prior work by mathematical role

Picard cohomology and the 2-primary alternatives. Swinnerton-Dyer identifies the algebraic Brauer quotient over an algebraic number field with Picard cohomology and proves that its possible nonzero 2-primary parts on a cubic surface are $\mathbf{Z}/2$ and $(\mathbf{Z}/2)^2$ [5]. Elsenhans and Jahnel place the corresponding Galois images inside U_1 and U_3 , respectively, and prove the restriction and class-map statements [2]. These source theorems, not the selected subgroup computation, supply the universal quantifier in theorem 3.2.

Orientation and cyclic algebras. The index-two orientation field and the cyclic algebra obtained from a norm divisor are prior mechanisms [2]. Explicit descent, including a full- U_1 example with line orbits [12, 15], also predates this instance [1]. What is computed here is the orientation square and canonical norm-divisor quartic for equation (3).

Resolvers and moduli. A degree-36 double-six resolver is already explicit [3], and the associated moduli covers occur in resolvent-degree theory [4]. We claim no generic resolver construction; the resolver here selects an exact subfield of K .

The number-field bridge. The subgroup statements concern only finite subgroups of $W(E_6)$, the integral lattice, and line configurations, while Swinnerton-Dyer works over algebraic number fields. We construct the norm divisor directly over F_D and use the number-field Hochschild–Serre sequence, for which Viray is a modern supporting locator [6].

2.4 Source, machine, and written-bridge contract

Source theorems control every possible image, exact arithmetic verifies this instance, and written algebraic geometry connects them to fields, divisors, and Brauer classes. Their division of labor is recorded in table 1.

The exact tuple used below has the intentionally layer-specific machine status

Claim	Source theorem	Exact instance input	Written bridge
Universal degree	Complete $\mathbf{Z}/2-U_1$, $(\mathbf{Z}/2)^2-U_3$ alternatives	Full $W(E_6)$ line field	$K^{H_L} = K \cap L$, index factors, tower law
Equality fields	Same classification	U_1 index, core, normalizer, and 36 configurations	Equality of indices and Galois correspondence
Brauer jump	Number-field Picard/Brauer identification	Integral $H^1(W(E_6), \Lambda)$ and $H^1(U_1, \Lambda)$	Inflation and Hochschild–Serre
Exact fields	General resolver/descent context only	Resolver coefficients, irreducibility, stabilizers	Orbit–stabilizer and fixed fields
Quartic	None	Gauge, 60×31 matrix, nonzero pivot, restrictions	Hilbert–90 descent and ambient lifting
Quaternion	General double-six class mechanism	Orientation, line distinctness, Picard action	Degree exhaustion, norm-divisor criterion, cocycle comparison

Table 1: Every theorem claim is split into its source, exact-instance, and written components. No column silently certifies another.

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Paper and full-project identities are separate external records, so the machine envelope is never cited as its own release provenance.

3 The universal minimum-degree theorem

This section proves the part of theorem 1.1 that quantifies over every finite extension $L/\mathbf{Q}$. The exact U_1 calculation enters only after the complete source theorem has reduced the possibilities to two branches.

Lemma 3.1 (Intersection fixed field). *Let $L/\mathbf{Q}$ be finite, put $G_L = \text{Gal}(\overline{\mathbf{Q}}/L)$, and let*

$$H_L = \text{im}(G_L \longrightarrow \text{Gal}(K/\mathbf{Q})). \quad (10)$$

Then

$$K^{H_L} = K \cap L, \quad [K \cap L : \mathbf{Q}] = [W(E_6) : H_L].$$

Proof. Because $K/\mathbf{Q}$ is finite Galois, restriction identifies the image of G_L with $\text{Gal}(K/K \cap L)$. Its fixed field in K is therefore $K \cap L$. The degree formula follows from the Galois correspondence and equation (4). $\square$

Theorem 3.2 (Sharp degree threshold). *If $L/\mathbf{Q}$ is finite and $(\text{Br}(Y_L)/\text{im Br}(L))[2] \neq 0$, then*

$$36 \mid [K \cap L : \mathbf{Q}] \mid [L : \mathbf{Q}]. \quad (11)$$

If $[L : \mathbf{Q}] = 36$ and $(\text{Br}(Y_L)/\text{im Br}(L))[2] \neq 0$, then

$$L = K \cap L = K^{wU_1w^{-1}} \quad (12)$$

for some $w \in W(E_6)$, with a unique double-six embedded in the fixed algebraic closure. Conversely, each field $K^{wU_1w^{-1}}$ has degree 36, and its Brauer quotient is nonzero.

Proof. Since $Y_{\overline{\mathbf{Q}}}$ is rational, $\text{Br}(Y_{\overline{\mathbf{Q}}}) = 0$. The inflation and number-field Hochschild–Serre identifications in lemma 5.1 and equation (33), applied with $k = L$, turn the hypothesis into

$$H^1(H_L, \Lambda)[2] \neq 0.$$

2-primary quotient	forced containment	ambient index
$\mathbf{Z}/2$	$H_L \subseteq wU_1w^{-1}$	$[W(E_6) : U_1] = 36$
$(\mathbf{Z}/2)^2$	$H_L \subseteq wU_3w^{-1}$	$[W(E_6) : U_3] = 720 = 36 \cdot 20$

Table 2: Both complete classification branches force divisibility by 36, but only the U_1 branch can attain index 36.

The complete 2-primary classification [5, 2] gives exactly the two containments in table 2, for a suitable $w \in W(E_6)$.

In the first branch, the subgroup index factors as

$$[W(E_6) : H_L] = 36[U_1 : w^{-1}H_Lw]. \quad (13)$$

In the second,

$$[W(E_6) : H_L] = 720[U_3 : w^{-1}H_Lw]. \quad (14)$$

Thus $36 \mid [W(E_6) : H_L]$ in either case. By lemma 3.1, this is the first divisibility in equation (11). The inclusion $K \cap L \subseteq L$ and the tower law give the second.

Now suppose, in addition, that $[L : \mathbf{Q}] = 36$. Every factor in equation (11) is one, so $L = K \cap L$ and $[W(E_6) : H_L] = 36$. The index-720 branch is impossible. Equation (13) forces $H_L = wU_1w^{-1}$, and the fixed-field statement in equation (12) follows.

The exact configuration action verifies that U_1 is self-normalizing and is the stabilizer of exactly one double-six. Therefore its conjugates are indexed by

$$[W(E_6) : N_{W(E_6)}(U_1)] = [W(E_6) : U_1] = 36,$$

and they correspond to the thirty-six double-sixes. Distinct conjugate subgroups have distinct fixed fields by Galois correspondence. This proves uniqueness of the embedded field attached to D . The fields are permuted transitively by $W(E_6) = \text{Gal}(K/\mathbf{Q})$, hence form a single $\mathbf{Q}$ -isomorphism type.

Conversely, $K^{wU_1w^{-1}}$ has degree 36. The exact integral calculation $H^1(U_1, \Lambda) \cong \mathbf{Z}/2$, together with the arithmetic bridge proved in section 5, shows that its Brauer quotient is nonzero. $\square$

Corollary 3.3 (Sharp threshold). *The smallest degree of a finite extension $L/\mathbf{Q}$ for which $(\text{Br}(Y_L)/\text{im Br}(L))[2] \neq 0$ is 36. Every such degree is divisible by 36.*

Proof. The divisibility is equation (11); existence in degree 36 is the converse part of theorem 3.2. $\square$

Remark 3.4 (Why one double-six is insufficient). The existence of a selected U_1 -fixed configuration proves attainment, not universal minimality. Without the $(\mathbf{Z}/2)^2$ - U_3 row in table 2, the proof would leave an uncontrolled family of Galois images. The fact that 720 is itself divisible by 36 is what allows the two branches to yield one sharp divisibility theorem.

4 Exact double-six and orientation fields

The degree theorem identifies the subgroup that must occur at equality. We now identify its fixed field inside the particular extension $K/\mathbf{Q}$. The construction begins with characteristic-zero incidence, passes to two separating orbit invariants, and ends with a degree-12 carrier for the selected lines.

4.1 The exact configuration set

On the complete Grassmann chart used to construct E , let $\tilde{g}(d) \in \mathbf{Z}[d]$ be the raw primitive degree-27 eliminant, let $c_g = [d^{27}]\tilde{g} \neq 0$, and put

$$g(d) = c_g^{-1}\tilde{g}(d). \quad (15)$$

A geometric line is recovered from a root $x = d$ of the monic eliminant g . The remaining line coordinates are rational polynomial functions of x . For roots x, y , exact divided differences give a polynomial $J(x, y)$ whose vanishing is equivalent to incidence of the corresponding lines. In the algebra $(\mathbf{Q}[x]/(g(x)))[y]$, put

$$H_x(y) = \gcd_y(g(y), J(x, y)). \quad (16)$$

The characteristic-zero identities are

$$\deg_y H_x = 10, \quad \gcd(H_x, y - x) = 1, \quad g(y) = H_x(y)Q_x(y), \quad \deg_y Q_x = 17. \quad (17)$$

Thus every line meets exactly ten other lines, with no diagonal counted. The number of unordered meeting pairs is $27 \cdot 10/2 = 135$. Exact independent set enumeration then gives

$$135 \text{ meeting pairs}, \quad 72 \text{ sixers}, \quad 36 \text{ double-sixes}. \quad (18)$$

Good-prime graphs provide label-compatible cross-checks, while equation (17) is the characteristic-zero authority; no ordering of approximate roots enters equation (18).

The exact $W(E_6)$ -action on these configurations has stabilizer data

$$|U_1| = 1440, \quad [W(E_6) : U_1] = 36, \quad \text{core}_{W(E_6)}(U_1) = 1, \quad N_{W(E_6)}(U_1) = U_1. \quad (19)$$

Its action on the twenty-seven lines has orbits of sizes 12 and 15, and U_1 fixes exactly the selected double-six.

4.2 Separating invariants and resolvers

Let d_i be the corresponding root of g , and define the fixed integral scaled coordinate

$$\alpha_i = c_g d_i. \quad (20)$$

For an unordered double-six D , define

$$\theta_D = \sum_{\ell_i \in D} \alpha_i. \quad (21)$$

For an orientation $D = (\mathcal{E}, \mathcal{G})$, define

$$\beta_D = \sum_{\ell_i \in \mathcal{E}} \alpha_i - \sum_{\ell_i \in \mathcal{G}} \alpha_i, \quad \delta_D = \beta_D^2. \quad (22)$$

The orbit products

$$R_\theta(T) = \prod_D (T - \theta_D), \quad R_\delta(T) = \prod_D (T - \delta_D) \quad (23)$$

belong to $\mathbf{Q}[T]$, are separable and irreducible, and have degree 36. Candidate-blind Chinese remaindering, with proved coefficient bounds, is recorded in appendix B. At two good reductions the factor-degree patterns are

$$[1, 5, 5, 5, 10, 10], \quad [9, 9, 9, 9]. \quad (24)$$

invariant	stabilizer	orbit size	generated field	role
θ_D	U_1	36	F_D	carrier primitive
δ_D	U_1	36	F_D	orientation square/radicand
β_D	U_1^+	72	F_D'	oriented sixer

Table 3: Exact orbit invariants, their stabilizers, and fixed fields.

Their proper subset-sum degree sets are disjoint, whereas a rational factor would give a degree in both. Hence the resolvers are irreducible.

The action on the exact orbit values gives

$$\text{Stab}(\theta_D) = \text{Stab}(\delta_D) = U_1, \quad \text{Stab}(\beta_D) = U_1^+. \quad (25)$$

The 36 values δ_D are distinct, the 36 values β_D are nonzero, and all 72 oriented values are separated exactly. The central involution $\iota \in U_1 \setminus U_1^+$ sends β_D to $-\beta_D$.

Theorem 4.1 (Configuration and orientation fields). *For every double-six D ,*

$$\mathbf{Q}(\theta_D) = F_D = \mathbf{Q}(\delta_D), \quad F_D' = F_D(\beta_D) = F_D(\sqrt{\delta_D}). \quad (26)$$

Both resolvers in equation (23) have normal closure K .

Proof. The stabilizers in equation (25) and the Galois correspondence give

$$\mathbf{Q}(\theta_D) = K^{U_1} = \mathbf{Q}(\delta_D).$$

Their orbit sizes are 36, so the degree-36 orbit polynomials are the minimal polynomials. The kernel of the coset action on the roots is $\text{core}_{W(E_6)}(U_1)$, which is trivial by equation (19). Hence the splitting field of either polynomial is the full K .

The same argument for β_D gives $\mathbf{Q}(\beta_D) = K^{U_1^+}$. Since U_1^+ has index two in U_1 and $\beta_D^2 = \delta_D$, this field is exactly the quadratic extension in equation (26). $\square$

Table 3 summarizes the three invariants. The equality $\mathbf{Q}(\theta_D) = \mathbf{Q}(\delta_D)$ is a labeled stabilizer theorem. We neither need nor certify an expanded polynomial identity $\delta_D = P(\theta_D)$.

4.3 The twelve-line carrier

Let $d_1, \dots, d_{12}$ be the line parameters in D , and let the remaining fifteen roots of g be $d_{13}, \dots, d_{27}$. The orbit decomposition [12, 15] gives monic polynomials

$$A_{12}(d) = \prod_{i=1}^{12} (d - d_i) \in F_D[d], \quad B_{15}(d) = \prod_{i=13}^{27} (d - d_i) \in F_D[d]. \quad (27)$$

By equations (20) and (21), its subtop coefficient is

$$[d^{11}]A_{12} = -\sum_{i=1}^{12} d_i = -\theta_D/c_g. \quad (28)$$

With the fixed monic normalization of g , exact division and an independent forward product prove

$$g(d) = A_{12}(d)B_{15}(d). \quad (29)$$

The 13×36 coefficient table is reconstructed with certified CRT bounds. Exact division and an independent forward product agree in all 28 coefficients. Together with the orbit binding, this identity—not the stored table alone—certifies that A_{12} cuts out exactly the twelve lines of D ; see appendix B.

5 Picard cohomology and the Brauer jump

The preceding sections locate the smallest possible image subgroup. To prove that its fixed field actually attains the bound, we now pass from exact integral group cohomology to the arithmetic Brauer quotient. This passage is a written number-field argument, not a semantic label attached to a Smith normal form.

5.1 Integral cohomology

The exact Picard action is generated from the six standard E_6 reflections on the marked lattice Λ . Rebuilding cocycle relations and principal cocycles over $\mathbf{Z}$, and then computing Smith normal forms, gives

$$H^1(W(E_6), \Lambda) = 0, \quad H^1(U_1, \Lambda) \cong \mathbf{Z}/2. \quad (30)$$

The calculation is integral: it does not tensor with $\mathbf{Q}$, reduce modulo two, or infer torsion from a rational rank. The full Smith data and the independent cochain convention appear in appendix C.

Lemma 5.1 (Inflation for the line-field image). *Let $k/\mathbf{Q}$ be a number field, and suppose the G_k -action on Λ has finite image $H \subseteq W(E_6)$. Then*

$$H^1(k, \Lambda) \cong H^1(H, \Lambda).$$

Proof. Let N be the kernel of $G_k \rightarrow H$. It acts trivially on Λ . Hence

$$H^1(N, \Lambda) = \text{Hom}_{\text{cont}}(N, \Lambda).$$

A continuous image of the profinite group N in the discrete lattice Λ is finite, while the torsion-free group Λ has no nonzero finite subgroup. Thus $H^1(N, \Lambda) = 0$. The inflation–restriction sequence now makes inflation $H^1(H, \Lambda) \rightarrow H^1(k, \Lambda)$ an isomorphism. $\square$

5.2 The number-field Hochschild–Serre bridge

Over $\overline{\mathbf{Q}}$, a smooth cubic surface is rational, so

$$\text{Br}(Y_{\overline{\mathbf{Q}}}) = 0. \quad (31)$$

For a number field k , the low-degree Hochschild–Serre sequence has the segment

$$\text{Br}(Y_k)/\text{im Br}(k) \longrightarrow H^1(k, \text{Pic}(Y_{\overline{\mathbf{Q}}})) \longrightarrow H^3(k, \mathbf{G}_m). \quad (32)$$

The last group vanishes in this number-field setting. Because of equation (31), every Brauer class on Y_k is algebraic, and equation (32) yields

$$\text{Br}(Y_k)/\text{im Br}(k) \cong H^1(k, \Lambda). \quad (33)$$

This is the number-field form of the Picard/Brauer bridge used by Swinnerton-Dyer [5]; a modern account of the relevant low-degree sequence is given by Viray [6].

Theorem 5.2 (The degree-36 Brauer jump). *For the fixed cubic surface,*

$$\text{Br}(Y)/\text{im Br}(\mathbf{Q}) = 0, \quad (34)$$

whereas for every double-six field F_D ,

$$\text{Br}(Y_{F_D})/\text{im Br}(F_D) \cong \mathbf{Z}/2. \quad (35)$$

Proof. Over $\mathbf{Q}$, the image on Λ is the full group $W(E_6)$ by equation (4). Apply lemma 5.1 and equation (33) and the first equality in equation (30); this proves equation (34).

For $k = F_D = K^{U_1}$, the image of G_{F_D} on the lines and Picard lattice is U_1 . The same two bridges and the second equality in equation (30) give equation (35). Thus the field constructed in theorem 4.1 attains the lower bound of theorem 3.2. $\square$

Remark 5.3 (Cohomological scope). Equation (35) identifies an algebraic/cohomological quotient. It neither evaluates its nonzero element at a completion of F_D nor compares adelic points. The explicit global representative will be constructed in section 7; no local conclusion is implicit in the group isomorphism above.

6 The canonical determinant quartic

We next turn the twelve-line carrier into one canonical quartic. Two rank bounds have different origins. Geometry produces a section and hence an upper bound on the restriction rank; exact arithmetic supplies a nonzero pivot and hence the lower bound. Keeping both halves visible prevents a finite-field minor from being mistaken for an existence proof.

6.1 A proved quotient gauge

The space $H^0(\mathbf{P}_{F_D}^3, \mathcal{O}(4))$ has dimension 35. Since F is a nonzero cubic, multiplication by F embeds the four-dimensional space of linear forms:

$$F \cdot H^0(\mathbf{P}_{F_D}^3, \mathcal{O}(1)) \subset H^0(\mathbf{P}_{F_D}^3, \mathcal{O}(4)). \quad (36)$$

The quotient therefore has dimension 31.

Put

$$c = [u_0^3]F = 75081586157.$$

Using the source basis (Fu_0, Fu_1, Fu_2, Fu_3) and projecting to the coefficients of

$$u_0^4, \quad u_0^3u_1, \quad u_0^3u_2, \quad u_0^3u_3,$$

gives the triangular block

$$B_F = \begin{pmatrix} 75081586157 & 0 & 0 & 0 \\ -28576620789 & 75081586157 & 0 & 0 \\ -122000922135 & 0 & 75081586157 & 0 \\ -5364921951 & 0 & 0 & 75081586157 \end{pmatrix}. \quad (37)$$

Its determinant is

$$\det B_F = c^4 = 31778526453059635681033276764499400992765201 \neq 0. \quad (38)$$

Consequently the four coefficient functionals restrict isomorphically to the subspace in equation (36). Every class modulo F times a linear form has a unique representative in which those four coefficients vanish. This proves the gauge; deleting four monomials is not a convention made after inspecting the answer.

Let

$$\mathcal{M} = (m_0, m_1, \dots, m_{30})$$

be the remaining degree-four monomials in the locked descending lexicographic order, with

$$m_0 = u_0^2u_1^2. \quad (39)$$

The complete order appears in appendix D.

6.2 Restriction matrix

For each of the twelve carrier lines, restriction of

$$Q = \sum_{j=0}^{30} q_j m_j$$

is a binary quartic and supplies five scalar equations. We use the monic carrier in the d -variable with the scaling $\alpha_i = c_g d_i$ and subtop normalization of equations (20) and (28). Reducing the five coefficients against that degree-12 carrier gives the exact matrix

$$M_D \in \text{Mat}_{60 \times 31}(F_D), \quad M_D q = 0. \quad (40)$$

Rows are ordered first by binary degree $0, \dots, 4$ and then by carrier degree $0, \dots, 11$. The column order is $\mathcal{M}$.

Proposition 6.1 (Geometric upper bound). *The matrix M_D satisfies $\text{rank}_{F_D} M_D \leq 30$.*

Proof. By the marking fixed in equation (6), the divisor classes of the two sixers are

$$[\mathcal{E}] = e_\Sigma, \quad [\mathcal{G}] = 12h - 5e_\Sigma.$$

Hence

$$\mathcal{E} + \mathcal{G} \sim 12h - 4e_\Sigma = 4H_Y. \quad (41)$$

This geometric equivalence does not by itself provide an F_D -rational section, so we descend it explicitly.

Put $k = F_D$, and choose a k -rational hyperplane section $H_0 = \text{div}_Y(\ell)$. Both the unordered divisor $\mathcal{E} + \mathcal{G}$ and $4H_0$ are k -rational. Over $\bar{k}$, choose $r \in \bar{k}(Y)^\times$ such that

$$\text{div}(r) = \mathcal{E} + \mathcal{G} - 4H_0.$$

For $\sigma \in G_k$, the rational function $c_\sigma = \sigma(r)/r$ has zero divisor. Since $Y_{\bar{k}}$ is projective and integral, $c_\sigma \in \bar{k}^\times$. These scalars form a multiplicative cocycle. Hilbert theorem 90 gives $a \in \bar{k}^\times$ with

$$c_\sigma = \frac{\sigma(a)}{a}.$$

It follows that $r_0 = r/a$ lies in $k(Y)^\times$. The section

$$s_D = r_0 \ell^4 \in H^0(Y_k, \mathcal{O}_Y(4))$$

is defined over k and vanishes along $\mathcal{E} + \mathcal{G}$.

The hypersurface restriction sequence

$$0 \longrightarrow \mathcal{O}_{\mathbf{P}_k^3}(1) \xrightarrow{\cdot F} \mathcal{O}_{\mathbf{P}_k^3}(4) \longrightarrow \mathcal{O}_{Y_k}(4) \longrightarrow 0 \quad (42)$$

is surjective on global sections because $H^1(\mathbf{P}_k^3, \mathcal{O}(1)) = 0$. Thus s_D lifts to a nonzero ambient quartic class over k that restricts to zero on the twelve lines. After applying the unique gauge, this class is a nonzero vector in $\ker M_D$. Therefore $\text{rank } M_D \leq 30$. $\square$

Proposition 6.2 (Exact lower bound). *The matrix M_D has rank at least 30.*

Proof. Delete the m_0 -column and select the zero-based row set

$$\mathcal{R} = \{0, \dots, 10, 12, \dots, 20, 24, \dots, 29, 36, 37, 38, 48\}. \quad (43)$$

The resulting matrix N_D is 30×30 . Exact arithmetic at the denominator-good prime 7 computes the norm of its determinant as

$$\text{Norm}_{F_D/\mathbf{Q}}(\det N_D) \equiv 3 \pmod{7}.$$

Thus $\det N_D \neq 0$ in F_D . The replay verifies the same nonvanishing for all thirty-six conjugate configurations, and so $\text{rank } M_D \geq 30$. $\square$

quartic invariant	exact value
ambient monomials	35
subspace $F \cdot H^0(\mathcal{O}(1))$	4
gauge columns	31
restriction equations	$12 \cdot 5 = 60$
locked pivot size	30×30
restriction rank / kernel dimension	$30/1$
normalization	$[u_0^2 u_1^2] Q_D = 1$

Table 4: Dimension, gauge, rank, and normalization data for Q_D .

Theorem 6.3 (Determinant-defined quartic). *For every double-six D ,*

$$\text{rank}_{F_D} M_D = 30, \quad \dim_{F_D} \ker M_D = 1.$$

There is a unique kernel vector

$$q_D = (1, q_1, \dots, q_{30}),$$

and hence a unique normalized gauge representative

$$Q_D = \sum_{j=0}^{30} q_j m_j \tag{44}$$

whose restrictions to all twelve lines vanish. Every q_j is defined by Cramer's rule from the locked matrix N_D .

Proof. Propositions 6.1 and 6.2 give rank 30, hence a one-dimensional kernel. If a kernel vector had $q_0 = 0$, its entries in columns $1, \dots, 30$ would lie in the kernel of the invertible selected minor N_D , so the vector would be zero. Thus $q_0 \neq 0$ on the kernel line and can be uniquely normalized to one. Solving the selected equations by Cramer's rule gives equation (44); the independent replay then checks all 60 equations, not only the pivot rows, at all 36 conjugates. $\square$

At this stage we know that Q_D is a nonzero quartic class and vanishes on the twelve carrier lines. We have not yet promoted those restrictions to a divisor equality; that is a separate geometric argument in section 7.

7 Divisor exhaustion and the explicit Brauer generator

The linear system in section 6 only proves vanishing on twelve lines. We first show that those lines exhaust the quartic divisor. We then use the oriented quadratic field to express the divisor of a rational function as a norm. Finally, and separately, we compare the associated Picard cocycle with the unique nonzero cohomology class.

7.1 The exact divisor

Proposition 7.1 (Degree exhaustion). *The normalized determinant quartic satisfies*

$$\text{div}_{Y_D}(Q_D) = \mathcal{E} + \mathcal{G}. \tag{45}$$

Proof. The exact carrier and restriction replay supplies twelve distinct lines and shows that Q_D vanishes identically on each. Thus every line in $\mathcal{E} + \mathcal{G}$ is a prime component of $\text{div}(Q_D)$ with multiplicity at least one. The quartic is not F times a linear form: its gauge class is the

normalized nonzero vector in $\ker M_D$. It therefore defines a nonzero effective divisor on Y_{F_D} of class $4H_Y$.

Intersecting with a hyperplane gives

$$(4H_Y) \cdot H_Y = 4H_Y^2 = 4 \deg(Y) = 12. \quad (46)$$

Each of the twelve distinct carrier lines has hyperplane degree one, so the known components already contribute the full degree 12. Every nonzero effective residual divisor has positive hyperplane degree, and any multiplicity greater than one would also increase the degree. Neither is possible. Hence equation (45) holds exactly. $\square$

This argument is the geometric boundary between the last two machine gates: the 60×31 rank calculation constructs Q_D , while distinctness and equation (46) identify its divisor. Restriction vanishing alone would not rule out a residual curve.

7.2 The norm divisor and unramifiedness

Let

$$\ell = u_0, \quad \mathcal{L}_0 = \operatorname{div}_Y(\ell), \quad f_D = \frac{Q_D}{\ell^4}. \quad (47)$$

The exact line replay verifies that no carrier line is contained in the hyperplane $\ell = 0$. Over the oriented field F'_D , define

$$\mathcal{D} = \mathcal{E} - 2\mathcal{L}_0. \quad (48)$$

The nontrivial automorphism ι of F'_D/F_D exchanges $\mathcal{E}$ and $\mathcal{G}$, and fixes $\mathcal{L}_0$. Therefore

$$\begin{aligned} \operatorname{Norm}_{F'_D/F_D}(\mathcal{D}) &= \mathcal{D} + \iota(\mathcal{D}) \\ &= \mathcal{E} + \mathcal{G} - 4\mathcal{L}_0 \\ &= \operatorname{div}(f_D). \end{aligned} \quad (49)$$

The cyclic-algebra divisor criterion now shows that

$$\mathcal{A}_D = (F'_D/F_D, f_D) = (\delta_D, Q_D/u_0^4) \quad (50)$$

has zero residue at every codimension-one point of Y_{F_D} , and hence extends to an unramified class in $\operatorname{Br}(Y_{F_D})$. This is the same general double-six mechanism that appears in the prior class construction [2]; the input specific to this surface is the exact divisor equation (45).

7.3 Unramified does not yet mean nonzero

We identify the class of equation (50) in $H^1(U_1, \Lambda)$. In the basis of equation (5), put

$$d_0 = e_\Sigma - 2h. \quad (51)$$

The first sixer has class e_Σ , while its opposite has class $12h - 5e_\Sigma$. Since $[\mathcal{L}_0] = H_Y$,

$$[\mathcal{D}] = e_\Sigma - 2H_Y = 3e_\Sigma - 6h = 3d_0. \quad (52)$$

Write $U_1 = U_1^+ \times \langle \iota \rangle$, with ι the central involution exchanging the sixers. The exact Picard action gives

$$\iota(h) = 5h - 2e_\Sigma, \quad \iota(e_\Sigma) = 12h - 5e_\Sigma, \quad \iota(d_0) = -d_0. \quad (53)$$

Moreover,

$$\Lambda^{U_1^+} = \mathbf{Z}h \oplus \mathbf{Z}e_\Sigma. \quad (54)$$

For $x = ah + be_\Sigma$, direct substitution into equation (53) yields

$$(1 + \iota)x = (6a + 12b)h - (2a + 4b)e_\Sigma,$$

and

$$(\iota - 1)x = 4(a + 3b)h - 2(a + 3b)e_\Sigma.$$

Consequently

$$\ker(1 + \iota | \Lambda^{S_6}) = \mathbf{Z}d_0, \quad (\iota - 1)\Lambda^{S_6} = 2\mathbf{Z}d_0. \quad (55)$$

With the norm-divisor convention in equation (49), the Hochschild–Serre cocycle of $\mathcal{A}_D$ is zero on $U_1^+ = S_6$ and takes ι to

$$[\mathcal{D}] = 3d_0.$$

If this cocycle were a U_1 -coboundary, its vanishing on S_6 would force a cobounding class into Λ^{S_6} . Its value at ι would then lie in $(\iota - 1)\Lambda^{S_6} = 2\mathbf{Z}d_0$, contrary to

$$3d_0 \notin 2\mathbf{Z}d_0.$$

Thus the class is nonzero.

Theorem 7.2 (Explicit generator). *The quaternion*

$$\mathcal{A}_D = (\delta_D, Q_D/u_0^4)$$

is unramified on Y_{F_D} and represents the unique nonzero element of

$$\mathrm{Br}(Y_{F_D})/\mathrm{im}\,\mathrm{Br}(F_D) \cong \mathbf{Z}/2.$$

Proof. Unramifiedness follows from equation (49). The calculation equations (52) and (55) proves that its Picard cocycle is nonzero. By equations (30) and (33), this cocycle is the unique nonzero cohomology class and therefore the generator of the displayed Brauer quotient. $\square$

Remark 7.3 (Global boundary). The proof separates three logically different assertions: the norm divisor proves unramifiedness, the integral lattice proves nontriviality, and neither computes a local evaluation. In particular, theorem 7.2 does not assert that $Y(F_D)$ is empty or nonempty, and it does not establish a Brauer–Manin obstruction, a Hasse-principle failure, or a failure of weak approximation.

8 Exact replay, claim boundary, and limitations

The proof combines finite exact arithmetic with arguments that are not reducible to certificate booleans. This section records the boundary at theorem-relevant granularity. Full artifact identities and evidence hashes are deferred to appendix E.

8.1 Numbered gates and the separate scope firewall

Table 5 summarizes the eight mathematical gates. The final row is deliberately not a ninth gate: it is a fail-closed negative-scope firewall.

The independent checker calls no producer theorem helper. It executes G0–G7, reconstructs 492 payload and 28 schema leaves, and rejects 535 semantic/structural mutations plus nine malformed parser cases; the canonical suite has 33 tests. Five deterministic gzip witnesses are checked after decompression, since compressed digests alone are not authority. A self-excluding manifest binds 18 code files and 10 non-manifest results, for 29 live files including the manifest. The external-directory runner compares byte, size, mode, nanosecond-mtime, and inode snapshots. The sole official atomic refresh and mandatory nonmutating replay passed.

row	exact payload	inference used in the paper	not inferred
G0	frozen surface and line-field import	correct Y, E, K , full $W(E_6)$ premise	a new upstream line-field theorem
G1	characteristic-zero incidence and all-and-only configurations	counts 135, 72, 36 and exact double-six carrier	the universal 2-primary subgroup classification
G2	U_1, U_1^+ , Picard action, integral cohomology	selected subgroup facts and equation (30)	classification of every possible H_L
G3	exact degree-36 resolvers	coefficients, irreducibility, orbit binding	novelty of a general resolver construction
G4	orientation square and stabilizers	$F'_D = F_D(\sqrt{\delta_D})$	an expanded identity $\delta = P(\theta)$
G5	exact $A_{12}B_{15} = g$	the twelve-line carrier over F_D	any unrelated line-field conclusion
G6	gauge, pivot, all restrictions	rank 30 and canonical Q_D	divisor equality before degree exhaustion
G7	line distinctness, degree and norm inputs, Picard action	exact divisor, unramified nonzero generator with written bridges	local evaluation or a rational-point consequence
SCOPE	negative flags and hostile mutations	preservation of the stated claim boundary	a positive mathematical theorem

Table 5: Exact gate contract. G6 ends at the determinant quartic and rank; G7 contains divisor exhaustion, the norm/quaternion step, and nonzero-class matching.

The exact arithmetic does not prove the complete U_1/U_3 source theorem, the fixed-field identity in lemma 3.1, the number-field Hochschild–Serre bridge, the Hilbert–90 upper rank bound, the degree-exhaustion argument, or the comparison of the quaternion cocycle with $3d_0$. Those steps remain written in sections 3 and 5 to 7. Conversely, the cited general theorems do not verify a coefficient, carrier, minor, or restriction for equation (3).

8.2 Status and provenance boundary

The machine envelope has payload digest

72e498b544599dbb8c7c56b2fd999ed8be80bdb0abd19393b5e47ddf60ae4574

and certificate digest

3078baf167d2344982d9f93811f1fd59a8258c8178ecce4decdbd2b054b16092f.

The independent machine-layer status is `PASS_PREFREEZE_CODE_RESULTS`. It supports the exact premises but is not reused as a paper or full-project identity. Those larger identities are recorded externally after their own inventories are closed, avoiding a self-referential digest.

8.3 Data and code availability

The anonymous supplement contains this project under the stable relative directory

`henon_dynamics/henon_mu3_yukawa_minimal_brauer_jump/`.

The executable entryptpoint is `code/run_all.sh`; the certificate, check report, schema, five deterministic evidence archives, and self-excluding machine manifest are in `results/`. The companion surface and line-field theorem packages are present under the sibling relative directories

henon_mu3_rational_yukawa_surface/ and henon_mu3_yukawa_line_field/. The scoped manifest identifies the machine lane. The full project and paper identities are supplied externally because embedding either digest in the object it hashes would create a cycle.

8.4 Limitations and nonclaims

The theorem concerns only the global algebraic/cohomological quotient of equation (3). It supplies neither a generic resolver or classification nor a theorem for arbitrary cubic, Yukawa, or Hénon surfaces; it certifies neither $\delta = P(\theta)$ nor an expanded quartic table. The fields E, K, F_D, F'_D retain distinct roles, and no unrelated determinant field is identified with F'_D/F_D .

There is no claim about rational points, (stable) rationality, local evaluation, the Hasse principle, weak approximation, or a Brauer–Manin obstruction. Nor is there a complete bad-prime inertia or Picard–Artin package, conductor, Euler factor, root number, motive, variation of Hodge structure, automorphy, Calabi–Yau realization, dynamical theorem, Riemann-hypothesis statement, Hilbert–Pólya operator, or announcement of a later adaptive-sequence project.

9 Conclusion

For the fixed cubic surface equation (3), full $W(E_6)$ line arithmetic delays every nonzero 2-primary algebraic Brauer quotient until a degree divisible by 36:

$$36 \mid [K \cap L : \mathbf{Q}] \mid [L : \mathbf{Q}].$$

The equality fields are precisely the conjugate fixed fields of double-six stabilizers. Exact stabilizers identify $\mathbf{Q}(\theta_D) = \mathbf{Q}(\delta_D) = F_D$ and the oriented quadratic extension $F'_D = F_D(\sqrt{\delta_D})$. A Hilbert–90 rank sandwich and one exact pivot define the normalized quartic Q_D ; degree exhaustion gives $\text{div}(Q_D) = \mathcal{E} + \mathcal{G}$; and the class $[D] = 3d_0 \notin 2\mathbf{Z}d_0$ proves that $(\delta_D, Q_D/u_0^4)$ is the unique nonzero Brauer class over F_D .

The result moves from the certified full line-field action to the first possible cohomological jump for this one surface. It does not evaluate that class locally or decide rational points, rationality, or a Brauer–Manin obstruction. The exact code/results tuple is independently replayable from the anonymous supplement using the relative entrypoint described in section 8; paper and full-project identities remain separate from the self-excluding machine manifest.

A Primary-source ledger and the number-field bridge

The bibliography is deliberately restricted to the six entries audited before drafting. Table 6 records the precise result imported from each source and, equally important, the instance facts that the source does not supply.

Table 6: Primary-source locators and claim boundaries.

source	exact locator	use here	not supplied
[5]	p. 449; §2; Lemma 1, pp. 451–452; Lemma 2, p. 453; concluding classification, p. 458	number-field Picard/Brauer description; invariant-double-six criterion; complete nonzero 2-primary possibilities $\mathbf{Z}/2$ and $(\mathbf{Z}/2)^2$	the frozen resolvers, fixed fields, orientation square, quartic, or quaternion representative

source	exact locator	use here	not supplied
[2]	introduction §1.3; Lemma 2.5; Remark 3.3 and Notation 3.4; Theorems 3.5 and 4.4; §4.3; Definition 4.1; Proposition 5.8; Theorem 5.11; Fact 6.2, Remark 6.3, Corollary 6.4	U_1/U_3 containment and restriction; orientation; invariant double-six class map; cyclic/quaternion precedent	the degree-36 field or determinant quartic for equation (3); Corollary 6.4 is not used for a local evaluation
[1]	Theorem 6.6; Algorithm 6.7; Propositions 7.1 and 7.4; Example 9.2	explicit invariant-double-six descent, tritangent-plane context, sixer-splitting radicand, and a full- U_1 [12, 15] example	inversion of the present frozen full- $W(E_6)$ line field
[3]	Remark 4.34(iii), printed p. 22	prior-art boundary: a degree-36 resolver with roots corresponding to double-sixes	either resolver polynomial or the Brauer class in this paper
[4]	Theorem 5.6 and Corollary 5.9	moduli-cover and resolvent-degree context for double-sixes and ordered sixers	an arithmetic fixed field for the present surface
[6]	§2.1, p. 10, equation (24) and the following low-degree exact sequence	supporting Hochschild–Serre exposition	any instance computation or automatic Brauer–Manin conclusion

A.1 Why the subgroup theorem applies over F_D

The attaining base is the number field F_D , not $\mathbf{Q}$. Swinnerton-Dyer’s Picard/Brauer theorem already works over algebraic number fields [5]; the Elsenhans–Jahnel containment and restriction statements concern finite subgroups of $W(E_6)$, the Picard lattice, and line configurations [2]. Thus for any number field k they apply to

$$H = \text{im}(G_k \rightarrow W(E_6))$$

without a $\mathbf{Q}$ -specific step. For the representative, we construct $F'_D/F_D, \mathcal{D}, f_D$ in the present tower, prove $\text{div}(f_D) = \text{Norm}(\mathcal{D})$, calculate its Picard cocycle, and only then use equation (33).

A.2 Bounded novelty screen

The 2026-08-15 audit screened exact coefficients and title phrases, configuration/Brauer queries, and the sources’ identifiers. It supports only:

The bounded 2026-08-15 screen did not locate a prior exact computation of the degree-36 double-six field, its orientation square, and the determinant-defined quaternion generator for this frozen Yukawa surface.

This is not an absolute priority claim: the generic constructions remain prior art.

B Incidence, resolvers, and the twelve-line carrier

B.1 Characteristic-zero incidence formula

On the complete chart, a line has row matrix

$$\begin{pmatrix} 1 & 0 & a & b \\ 0 & 1 & c & d \end{pmatrix}$$

and points $(s, t, as + ct, bs + dt)$. The frozen lexicographic shape writes

$$a(x) = -\frac{h_a(x)}{A_a}, \quad b(x) = -\frac{h_b(x)}{A_b}, \quad c(x) = -\frac{h_c(x)}{A_c}, \quad d(x) = x,$$

where the three denominators are nonzero constants. Here ($x=d$), the raw primitive eliminant is $\tilde{g}(d)$, its leading coefficient is c_g , and the monic eliminant and integral orbit coordinate are $g = c_g^{-1}\tilde{g}$ and $\alpha_i = c_g d_i$, respectively, as in equations (15) and (20). Two chart lines at $x \neq y$ meet precisely when

$$(a(y) - a(x))(d(y) - d(x)) - (b(y) - b(x))(c(y) - c(x)) = 0. \quad (56)$$

For $\nu \in \{a, b, c\}$, put

$$D_\nu(x, y) = \frac{h_\nu(y) - h_\nu(x)}{y - x}.$$

After dividing equation (56) by $(y - x)^2$ and clearing the constant denominators, the exact incidence carrier is

$$J(x, y) = -D_a A_b A_c - D_b D_c A_a. \quad (57)$$

The polynomial J remains defined on the diagonal. Its gcd has no diagonal root and satisfies equation (17). At each good specialization all 27 copies of H_x equal the monic modular gcd. Specialization gives the upper degree bound 10, while the exact factor gives the lower bound.

The sixers are enumerated as six-element independent sets in the meeting graph. For a sixer S , its opposite is

$$S^\vee = \{\ell \notin S : \ell \text{ meets exactly five lines in } S\}.$$

Every exact sixer has $|S^\vee| = 6$, and pairing S with $S^\vee$ produces the 36 double-sixes in equation (18).

B.2 Resolver reconstruction and irreducibility

Resolver arrays run from constant to leading term and are monic. Every CRT prime is proved prime; the eliminant remains squarefree, denominators are units, and the modular orbit product is multiplied back. Signed reconstruction requires a modulus exceeding twice the coefficient bound.

At a squarefree good reduction, the degree of any rational factor is a proper subset sum of the modular factor degrees. The two sets arising from equation (24) are disjoint, proving irreducibility of both resolvers. Separately, the exact configuration action binds their roots all-and-only to the 36 double-sixes.

B.3 Carrier identity over $\mathbf{Q}(\theta_D)$

Write

$$A_{12}(d) = d^{12} + a_{11}d^{11} + \cdots + a_0, \quad a_i \in \mathbf{Q}(\theta_D).$$

object	reconstruction witness	independent identity
incidence H_x, Q_x	characteristic-zero monic degrees 10, 17; good primes 7, 37, $10^{50} + 12477$	$g = H_x Q_x$, $H_x \mid J$, no diagonal, all modular graphs agree
R_θ	99 primes; 4951-digit CRT modulus	orbit product, exact factors, incompatible subset sums
R_δ	198 primes; 9901-digit modulus exceeding the 9858-digit height requirement	orbit product, exact factors, incompatible subset sums
A_{12}	1048 primes; 100609-digit modulus; 13×36 coefficient table	28 zero division remainders and 28 matching forward coefficients

Table 7: Compact reconstruction ledger. Large arrays are kept in exact electronic witnesses rather than printed as decimal tables.

The subtop coefficient has the exact normalization

$$[d^{11}]A_{12} = -\sum_{i=1}^{12} d_i = -\theta_D/c_g. \quad (58)$$

Exact long division by A_{12} yields B_{15} with zero remainder, and an independent convolution recomputes all 28 product coefficients. The $[12, 15]$ orbit partition binds the roots of A_{12} to D , not merely to an arbitrary degree-12 factor.

The stable carrier-table digest is

$$72d4aef5120926ec09904b08219cba7cf2b49323bd085d05274e5c17e1ed90a1.$$

This digest identifies the compact table but does not replace the division, forward-product, or orbit-binding checks.

C Integral cohomology and class matching

C.1 Cochain convention and Smith forms

For a finite group Γ acting on Λ , a crossed homomorphism is a map $z : \Gamma \rightarrow \Lambda$ satisfying

$$z(gh) = z(g) + g z(h).$$

After fixing the exact group generators, the values on those generators form an integral vector. Multiplication relations in the enumerated group give an integral relation matrix whose kernel is $Z^1(\Gamma, \Lambda)$. Principal cocycles are the image of

$$\Lambda \longrightarrow Z^1(\Gamma, \Lambda), \quad v \longmapsto (g \mapsto gv - v).$$

The checker rebuilds the group, both integer lattices, and the Smith normal form of their quotient; it does not accept a stored cohomology label.

For both $\Gamma = W(E_6)$ and $\Gamma = U_1$, the relation matrix has rank 36 and the cocycle kernel has rank 6. The quotient data are

$$\begin{array}{c|c|c} \Gamma & \text{Smith diagonal} & H^1(\Gamma, \Lambda) \\ \hline W(E_6) & (1, 1, 1, 1, 1, 1) & 0 \\ U_1 & (1, 1, 1, 1, 1, 2) & \mathbf{Z}/2. \end{array} \quad (59)$$

All matrices use the marked basis $(h, e_1, \dots, e_6)$. The result is unchanged under conjugating U_1 , so every embedded double-six field has the same cohomological jump.

C.2 The S_6 -fixed plane and the central involution

The exact S_6 -invariant basis is

$$h = (1, 0, 0, 0, 0, 0, 0), \quad e_\Sigma = (0, 1, 1, 1, 1, 1, 1).$$

In the ordered basis (h, e_Σ) , the central sixer swap acts by

$$\iota(h) = 5h - 2e_\Sigma, \quad \iota(e_\Sigma) = 12h - 5e_\Sigma.$$

For completeness, if $x = ah + be_\Sigma$, then

$$(1 + \iota)x = 6(a + 2b)h - 2(a + 2b)e_\Sigma = 2(a + 2b)H_Y,$$

where $d_0 = e_\Sigma - 2h$. Hence its kernel is obtained from $a + 2b = 0$:

$$\ker(1 + \iota \mid \Lambda^{S_6}) = \mathbf{Z}d_0.$$

Similarly,

$$(\iota - 1)x = 4(a + 3b)h - 2(a + 3b)e_\Sigma = -2(a + 3b)d_0,$$

and therefore

$$(\iota - 1)\Lambda^{S_6} = 2\mathbf{Z}d_0.$$

These formulas are an independent two-dimensional check on the $\mathbf{Z}/2$ factor in equation (59).

C.3 Cocycle carried by the quaternion

Let $\chi : U_1 \rightarrow U_1/U_1^+ \cong C_2$ be the orientation character. The norm-divisor equality equation (49) defines a cyclic class whose Picard cocycle vanishes on $\ker \chi = U_1^+$ and, with the fixed divisor convention, has

$$z(\iota) = [\mathcal{D}].$$

The marked line classes give

$$[\mathcal{E}] = e_\Sigma, \quad [\mathcal{L}_0] = 3h - e_\Sigma,$$

so

$$z(\iota) = [\mathcal{D}] = e_\Sigma - 2(3h - e_\Sigma) = 3(e_\Sigma - 2h) = 3d_0.$$

This is a cocycle because $(1 + \iota)3d_0 = 0$. If it were principal, its zero restriction to S_6 would force the cobounding vector to lie in Λ^{S_6} , and its value at ι would lie in $2\mathbf{Z}d_0$. Since $3d_0 \notin 2\mathbf{Z}d_0$, the class is nonzero. The last line of equation (59) then shows that it is the unique nonzero element.

This calculation is the class-map bridge used in theorem 7.2. The norm-divisor identity is necessary for unramifiedness, but the parity of $3d_0$ against the coboundary lattice is what proves nontriviality.

D Determinant specification of Q_D

This appendix fixes every convention needed to recover the quartic without printing its expanded degree-36 coefficient table.

D.1 Gauge and monomial order

List all exponent vectors

$$(e_0, e_1, e_2, e_3), \quad e_i \geq 0, \quad e_0 + e_1 + e_2 + e_3 = 4,$$

in descending lexicographic order. Delete

$$(4, 0, 0, 0), \quad (3, 1, 0, 0), \quad (3, 0, 1, 0), \quad (3, 0, 0, 1).$$

The validity and uniqueness of this deletion are proved by the gauge block equations (37) and (38). The remaining ordered monomials are

$$\begin{array}{lll} m_0 = u_0^2 u_1^2, & m_1 = u_0^2 u_1 u_2, & m_2 = u_0^2 u_1 u_3, \\ m_3 = u_0^2 u_2^2, & m_4 = u_0^2 u_2 u_3, & m_5 = u_0^2 u_3^2, \\ m_6 = u_0 u_1^3, & m_7 = u_0 u_1^2 u_2, & m_8 = u_0 u_1^2 u_3, \\ m_9 = u_0 u_1 u_2^2, & m_{10} = u_0 u_1 u_2 u_3, & m_{11} = u_0 u_1 u_3^2, \\ m_{12} = u_0 u_2^3, & m_{13} = u_0 u_2^2 u_3, & m_{14} = u_0 u_2 u_3^2, \\ m_{15} = u_0 u_3^3, & m_{16} = u_1^4, & m_{17} = u_1^3 u_2, \\ m_{18} = u_1^3 u_3, & m_{19} = u_1^2 u_2^2, & m_{20} = u_1^2 u_2 u_3, \\ m_{21} = u_1^2 u_3^2, & m_{22} = u_1 u_2^3, & m_{23} = u_1 u_2^2 u_3, \\ m_{24} = u_1 u_2 u_3^2, & m_{25} = u_1 u_3^3, & m_{26} = u_2^4, \\ m_{27} = u_2^3 u_3, & m_{28} = u_2^2 u_3^2, & m_{29} = u_2 u_3^3, \\ m_{30} = u_3^4. \end{array}$$

The normalization is $q_0 = 1$; equivalently, $[u_0^2 u_1^2] Q_D = 1$.

D.2 The 60×31 matrix

For a carrier root d , write the corresponding line as

$$(s, t, a(d)s + c(d)t, b(d)s + dt).$$

For every m_j , expand

$$m_j(s, t, a(d)s + c(d)t, b(d)s + dt) = \sum_{\tau=0}^4 r_{\tau j}(d) s^{4-\tau} t^\tau. \quad (60)$$

Reduce $r_{\tau j}(d)$ modulo the monic carrier $A_{12}(d)$:

$$r_{\tau j}(d) \equiv \sum_{\rho=0}^{11} M_{12\tau+\rho, j} d^\rho \pmod{A_{12}(d)}.$$

This fixes the row convention

$$\text{row} = 12(\text{binary degree}) + (\text{carrier degree}), \quad (61)$$

with both degrees increasing. The matrix

$$M_D = (M_{rj})_{\substack{0 \leq r < 60 \\ 0 \leq j < 31}}$$

is therefore determined by the surface, chart shape, carrier, gauge, and orders above.

D.3 Pivot and Cramer’s rule

Let

$$\mathcal{R} = \{0, 1, \dots, 10, 12, \dots, 20, 24, \dots, 29, 36, 37, 38, 48\}.$$

Define

$$N_D = (M_{rj})_{\substack{r \in \mathcal{R}, \\ 1 \leq j \leq 30}}, \quad b_D = (M_{r0})_{r \in \mathcal{R}}.$$

The exact pivot certificate proves $\det N_D \neq 0$. If $N_D^{(j)}(b_D)$ denotes N_D with its j -th column replaced by b_D , then

$$q_j = -\frac{\det N_D^{(j)}(b_D)}{\det N_D}, \quad 1 \leq j \leq 30. \quad (62)$$

Equations (44) and (62), together with the monomial list, define Q_D exactly.

The pivot determinant has nonzero norm 3 modulo 7. The exact orbit replay checks all 36 conjugate determinant values and all $36 \cdot 60 = 2160$ restriction equations. Stable internal fingerprints are

pivot determinant coefficient digest:
1af5bfdc9b2f945094835fd81281305fb84dfc8208cb542874f2803420cd3a9e
canonical solution digest:
eb9e803e16f5623647843dd7636fe3d511af60f8f31c453ea6826cd3e4d25573

The fingerprints audit the exact arrays. They are not mathematical substitutes for gauge invertibility, the Hilbert–90 upper bound, pivot nonvanishing, or the complete restriction replay.

E Machine certificate and fail-closed contract

The identities below bind the exact machine input. Paper and full-project identities are external, non-self-referential records.

E.1 Envelope and upstream locks

object	SHA-256	bytes
certificate	3078baf167d2344982d9f93811f1fd59a8258c8178ecce4decdbd2b054b16092f	27326
independent check report	fb0afb77f130fb2d0a792af8d949e5c1a8e1b7864525dd62f1d1a41d99a79bcf	2176
schema file	81942b5f8012071cbc1bd24f8a85e04a2e7fea10b3f5cb8e8ca3014c6e822f72	1040
scoped manifest	864c05b18e0bdcafbcb5b5e3206840a1b25afa355b9737ebcc9d1806e33fcec5d	4855

Table 8: Authoritative live envelope. The manifest is self-excluding.

The canonical payload digest is

72e498b544599dbb8c7c56b2fd999ed8be80bdb0abd19393b5e47ddf60ae4574,

the canonical schema digest is

4c4675b71556867a5699574b5f3f54aa40551c76a14e22ac21c62370add61cc4,

and the independent replay-summary digest is

17b383c762b20f45767c4daffc704cbfd2f6e9a54f97886de641d11018b3ac68.

The envelope is `PREFREEZE_CODE_RESULTS_PASS`; this machine-layer status is intentionally distinct from the paper and full-project identities.

The upstream import uses the review-safe logical objects I56, P56, and R56 from the companion anonymous-supplement package `henon_mu3_yukawa_line_field`. Their exact object names are rebound inside the machine certificate rather than printed as public-repository locators here. The import replays the upstream certificate

26739ce5aedb4a3467645f9c1b2036d4d3eec9ce4d0dbce23d67ea7b67e5fbc4.

It also binds the upstream payload, check report, manifest, and Route, while preserving $E \neq K$.

E.2 Semantic evidence artifacts

artifact	compressed SHA-256 and bytes	decompressed SHA-256 and bytes
a12_crt_transcript.json.gz	5171ed975096c1cfffed221e7ecec7a87d3afd2970ccd91c431b5283587f7a22f18797009 bytes	1189d4b015b2024fede11fbc4361575eccbddeed9c3ff6043cd34d16ec311fb240146042 bytes
a12_table.json.gz	1034828fcfaad5262ea0388762947e78bf51f3986be642643dfd5f66711e3f615817606 bytes	9dd43ebbdd61873dae3f6437c160a0dfbc389934a62a38057b915955c117c3cc33700083 bytes
delta_crt.json.gz	4deead9914f31b0012afd91088339793330874a3b5156ceaeb1371fcb495f68597372 bytes	61ead9febb5ee8295c75980b81ba1c73c2d9cdae9e87dd0d0e76da899999a9288384 bytes
incidence_char0_witness.json.gz	4853641b143d3d7d0c2086fbee13f9d1f191bac960a989e8e85aede117cf80601697390 bytes	2c42ac21f43e54870b030c71facff31b0b0b5a05da544b7455f960e47448a3924297007 bytes
theta_crt.json.gz	91181a525e0acb17e73d2e96fd4e7d5d7a25913784ef8ad9d3be59c430a4fadd50282 bytes	5760dd3f4a1e07834f974e340f6cd488d9b793dab8efa0864505903cf9e1bcb3132705 bytes

Table 9: The five large evidence inputs. Each has gzip mtime zero and must survive deterministic decompression and semantic replay.

E.3 Gate and executable separation

The independent check report binds the gate payloads:

```
G0 7b4b72505f6480ae085a5b8e27749c28c9fab87e26c883b1d74da33f5baa7fc8
G1 a25b718a28f5260b7cbfa1c5ee954f10de7f8b586b0cd573ac33038b49d90d58
G2 a462027bf02dac1f1d54a373130d11e262db76247821feffb7143accb88af3e9
G3 323f89c767f44a705009a05b0eaecc2804a1201ff46ab52949eabb1582f9fea1
G4 f07c2ed52daddb00428e016bc17c1e299df218553b58aca725f0254f171f328c
G5 fce3d88a04abdf2b6c3e6ebd76a7776d4e14daaba048ffa74b76276d4f367e32
G6 3f5d2802efb69942dd79e74e9b3c5aae47091b04223b90dda08183880637245b
G7 ea63a74a6db2024f0cc937dc90675986f122659e2decf629ad2020dcdb6d798
```

The producer, independent checker, pipeline, atomic promoter, manifest builder, runner, and test-suite digests are respectively

```
30b6aae77ceaa33fb2adf2d0ec69477afe2a8b11c789e4dce857873f46e52941,
a44ad192b7840bba4b4a74899de0d369e79f3a8d41853a84c854ed34739e0064,
00dfc9e376525c16bb0925f1fc7af2a0b7adef3945e4f3020fbba0fd900faf0e,
c2057c74e4973576a3e67b6ef27a1da261ff7263480e91342c91da04d1a32558,
822e291876ca6df1822fbb2a8930fd8fd842d7bf4e58a2c95307f7e0036a4a8b,
8bcb6f206c991c827b8af63bedfd63b7991d9f376cfdb969ff2333e88fbfc6f4,
e883b79c1e104054526155ccc5c459b430779a8a16779cec4d4ec082d91eb591.
```

The manifest binds 18 code files and 10 non-manifest results; the live inventory adds the manifest.

E.4 Fail-closed behavior

The suite mutates every classified scalar leaf while rebinding exposed hashes, and independently rejects semantic changes. It also tests malformed parsing, optimized execution, import-path

injection, deterministic gzip, special links/files, stale output, lock replacement, and hostile shell startup. Failure injection into the fixed nine-file refresh must restore hash, size, mode, and nanosecond mtime. The nonmutating default runner compares byte/mode/mtime/inode snapshots from an external directory and retains uncertain stages fail-closed. These checks protect evidence identity; they do not prove the source classification, Hilbert–90 descent, divisor exhaustion, or cocycle comparison.

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